

Data Analytics Bootcamp

Module: Introduction to Data Analysis using Excel Foundation

Session: Basic Data and Statistical Concepts

- Data Literacy
- Statistical Foundations
- Descriptive Statistics
- Statistical Inference and Sampling Techniques
- Regression Analysis

Session: Data Cleaning, Preparation, and Management

- Data relationships, data shapes, and primary/unique keys and identifiers
- Basic data cleaning
- Data extraction from external sources
- Sorting, Filtering, and Merging Data
- Common excel formulas and functions

Session : Further Concepts in Data Cleaning

- Excel Functions for Data Cleaning (E.g. Left, Right, Concatenate, etc)
- Conditional Formatting
- Data Validation and Error Checking

Session : Basic Data Processing

- IF and Nested IF statements
- Absolute and Relative cell references

Session: Data Analysis with Intermediate Excel

- Processing large datasets
- Find, Find & Replace
- Advanced Commands (VLookUp, Index, Match)

Session: Introduction to Pivot Tables

- Navigating Pivot Tables Interface
- Creating basic Pivot Tables
- Field Settings and layout options
- Working with pivot table data

Session: Advance Data Analysis with Pivot Tables

- Advanced pivot table functions
- Data slicing
- Pivot charts and visualizations
- Advanced Pivot table visualizations

Session: Further Data Visualization on Excel

- Intermediate and Advanced Excel Charting Techniques
- Excel Add-ins for advanced visualization
- Data Analysis and Visualization on Google Sheets.

Module: Data Visualisation and Analysis using PowerBI**Session : Introduction to PowerBI/ Connecting & Shaping Data**

- Download and install Power BI Desktop, and adjust the settings.
- Understand the role that Power BI plays within the broader Microsoft ecosystem
- Explore core components of the Power BI Desktop interface
- Review the business intelligence workflow.
- Explore Power BI's query editor and understand the role that Power Query plays in the larger BI workflow
- Introduce different types of connectors and connectivity modes available for getting data into Power BI
- Review tools for checking data quality and key profiling metrics like column distribution, empty values, errors, and outliers
- Transform tables using text, numerical and date/time tools, pivot and group records, and create new conditional columns
- Practice combining, modifying, and refreshing queries

Session: Data Modeling

- Understand the basic principles of data modeling, including normalization, fact & dimension tables, and common schemas

- Create table relationships using primary and foreign keys, and discuss different types of relationship cardinality
- Configure report filters and trace filter context as it flows between related tables in the model
- Explore data modeling options like hierarchies, data categories, and hidden fields

Session: DAX

- Introduce DAX fundamentals and learn when to use calculated columns and measures
- Understand the difference between row context and filter context, and how they impact DAX calculations
- Learn DAX formula syntax, basic operators and common function categories (math, logical, text, date/time, filter, etc.)
- Explore nested functions, and more complex topics like iterators and time intelligence patterns

Session: Data Visualization

- Review frameworks and best practices for visualizing data and designing effective reports and dashboards
- Explore tools and techniques for inserting, formatting and filtering visuals in the Power BI Report view
- Add interactivity using tools like bookmarks, slicer panels, parameters, tooltips, and report navigation
- Learn how to configure row-level security with user roles
- Optimize reports for mobile viewing using custom layouts

Module : Database Management SQL

Session: Installing SQL Workbench

Session: Introduction to SQL

- Overview of Database Management Systems(DBMS)
- Basic SQL syntax and structure
- Flow of SQL commands
- Data Types in SQL

- Running SQL commands using SELECT statement
- Retrieving data from tables

Session: Filtering, Sorting, and Aggregating Data

- Using WHERE clause to filter data based on conditions
- Sorting query results using ORDER BY clause
- Limiting the number of results with LIMIT clause
- Using aggregate functions for summary statistics
- Grouping query results using GROUP BY clause
- Filtering grouped data with HAVING clause

Session: Table Joins and Case Statements

- Case statements
- Understanding relationships between tables
- Using joins to combine data from multiple tables

Session: Advanced Query Techniques

- Further practice with different types of joins
- Working with subqueries
- Working with derived tables
- Common table expressions (CTEs)

Session: Window Functions and Data Modification

- Working with window functions
- Ranking data using RANK and DENSE_RANK
- Modifying data in tables using ALTER, RENAME, INSERT, UPDATE, DELETE
- Using UNION, INTERSECT, and EXCEPT to combine query results

Session : Creating New Tables

- Further practice with window functions
- Creating new tables
- Inserting values into new tables
- Modifying and updating new tables

Session : Advanced Topics and Artificial Intelligence

- Working with Primary Keys
- Auto-Increments
- Updating Tables
- Indexing
- Using AI in SQL programming

Session : Date Variables and Artificial Intelligence

- Dealing with date variables in SQL data
- Setting variables in SQL
- Using AI in SQL programming

- Wrapping up
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Module: Spatial Data Analysis using GIS

Session: Introduction to Spatial Analysis

- Understanding Spatial Data: Definition, significance, and examples of spatial data.
- Spatial Data Types: Differentiating between vector and raster data.
- Applications of Spatial Analysis: Overview of how spatial analysis is used in various fields such as environmental science, urban planning, and public health.
- Introduction to GIS: Understanding Geographic Information Systems and their role in spatial analysis.

Session: Introduction to QGIS

- Getting Started with QGIS: Installing QGIS and familiarizing with the interface.
- Basic Operations: Opening and viewing spatial data, navigating the map canvas, and managing layers.
- Data Import and Export: How to import and export different spatial data formats in QGIS.

Session: Creating Shapefiles (Point, Line, and Polygon)

- Understanding Shapefiles: The structure and components of shapefiles.
- Creating New Shapefiles: Step-by-step guide to creating point, line, and polygon shapefiles.
- Editing Shapefiles: Adding, modifying, and deleting features in shapefiles.

Session: Learning Basic Cartography

- Map Elements: Understanding essential map elements (title, legend, scale, north arrow).
- Design Principles: Color theory, symbolization, and layout design for clear and effective map-making.
- Creating Maps: Practical exercises to design maps using QGIS.

Session: Learning Basic Vector Analysis Tools

- Spatial Queries: Selecting features based on spatial relationships (e.g., proximity, intersection).
- Buffer Analysis: Creating buffers around features and understanding their applications.
- Overlay Analysis: Performing spatial operations like intersect, union, and difference.

Session: Creating Heatmaps

- Understanding Heatmaps: Concept and applications of heatmaps in representing data density.
- Generating Heatmaps in QGIS: Step-by-step guide to creating heatmaps from point data.
- Customization: Adjusting parameters to refine the heatmap visualization.

Session: Using Google Maps to Create Maps in QGIS

- Integrating Google Maps: Methods to use Google Maps as basemaps in QGIS projects.
- Georeferencing: Aligning external map images with spatial data using georeferencing tools.
- Practical Exercise: Creating a map project in QGIS with Google Maps as the base layer.

Session: Introduction to Raster Data

- Understanding Raster Data: Definition, structure, and examples of raster data.
- Raster Analysis Tools: Introduction to basic raster operations (e.g., map algebra, reclassification).
- Working with Satellite Imagery: Basics of accessing and analyzing satellite imagery in QGIS.

Session: Real-world Examples of GIS in Various Industries and Fields

- Environmental Management: Use cases of GIS in conservation, pollution monitoring, and resource management.
- Urban Planning: Applications in zoning, infrastructure development, and traffic management.
- Public Health: Spatial analysis in epidemiology, access to healthcare facilities, and health outcome mapping.
- Agriculture: Precision farming, crop yield analysis, and soil mapping.

Module : Python for Data Analysis

Session 0: Installation

- Introduction to the Python programming language and its applications
- Setting up the Python environment: installation of Python and necessary libraries
- Configuring the development environment: IDEs, text editors, and Jupyter Notebook

Session : Python Basics

- Introduction to Python: history, features, and advantages
- Expressions and operators: arithmetic, assignment, comparison, and logical
- Understanding type() function and type inference
- Introduction to data structures: lists, tuples, and dictionaries

Session : Python Basics

- Recap of Python basics
- Working with arithmetic operators: addition, subtraction, multiplication, division, modulus, and exponentiation
- Using comparison operators: equal to, not equal to, greater than, less than, etc.
- Logical operators: and, or, and not
- Exploring advanced data types: sets and strings manipulation.

Session : Expressions, Conditional Statements & For Loop

- Evaluating expressions: operator precedence and associativity
- Introduction to conditional statements: if, elif, and else
- Executing code based on conditionals.
- Understanding the flow of control in conditional statements
- Iteration using the for loop: range(), iteration over lists, and strings.

Session : While loop, Break and Continue Statements, and Nested Loops

- Working with while loop: syntax, conditions, and examples
- Combining loops and conditionals
- Using the break statement to exit loops prematurely.
- Utilizing the continue statement to skip iterations.
- Implementing nested loops for complex iterations

Session : Functions

- Introduction to functions: purpose, advantages, and best practices

- Defining and calling user-defined functions
- Parameters and arguments: positional, keyword, and default values
- Return statement and function output.
- Variable scope and lifetime
- Function documentation and code readability

Session : Exception Handling and File Handling

- Understanding exceptions: errors, exceptions, and exception hierarchy
- Handling exceptions using try-except blocks: handling specific exceptions, multiple exceptions, and else and finally clauses.
- Raising exceptions and creating custom exception classes
- File handling in Python: opening, reading, writing, and closing files.
- Working with different file modes and file objects

Session : Python Modules: NumPy and Matplotlib

- Introduction to the NumPy module: features and applications
- Working with multidimensional arrays: creation, indexing, slicing, and reshaping
- Performing element-wise operations: arithmetic, logical, and statistical
- Overview of the Matplotlib module: data visualization and plotting
- Customizing plots: line properties, markers, colors, labels, and legends

Session : Mathematics for Data Science

- Vectors and Matrices: Definition, addition, scalar multiplication.
- Matrix Multiplication: Concept and application in data transformation and neural networks.
- Eigenvalues and Eigenvectors: Importance in dimensionality reduction techniques like PCA
- Derivatives and Gradients: Understanding how changes in input affect changes in output.
- Partial Derivatives and the Gradient Vector: Application in gradient descent and optimization.
- Introduction to Optimization: Concept of loss functions and how gradients are used to minimize them.
- Probability Theory: Basics and conditional probability.
- Bayes' Theorem: Importance in machine learning for making predictions.
- Descriptive Statistics: Mean, median, mode, variance, and standard deviation.
- Distributions: Normal distribution and its significance in machine learning.
- Beyond Gradient Descent: Introduction to stochastic gradient descent and mini-batch gradient descent.
- Regularization Techniques: L1 (Lasso) and L2 (Ridge) regularization to prevent overfitting.
- Introduction to Convex Optimization: Understanding convex functions and their relevance to machine learning optimization.

Module : Exploratory Data Analysis (EDA) and Machine Learning

Introduction and Missing Value Analysis

- Introduction to Exploratory Data Analysis (EDA)
- Importance of EDA in data analysis
- Steps involved in EDA
- Handling missing values: identification, analysis, and treatment strategies • Imputation techniques for missing values

Data Consistency, Binning, and Outlier Analysis

- Data consistency checks using fuzzy logic
- Binning and discretization techniques for continuous variables
- Outlier detection and analysis methods
- Handling outliers: techniques for treatment or removal

Feature Selection and Data Wrangling

- Importance of feature selection in EDA
- Feature selection techniques: filter methods, wrapper methods, and embedded methods
- Data wrangling: cleaning and transforming data for analysis
- Handling categorical variables: encoding techniques

Inference, Hypothesis Testing, and Visualization

- Inference and hypothesis testing in EDA
- Common statistical tests: t-test, chi-square test, ANOVA, etc.
- Visualization techniques for EDA: histograms, box plots, scatter plots, etc.
- Hands-on practical session for complete EDA using a dataset

Professional Development Series

1. Email Writing

- Basics of Professional Email Communication: Structure, tone, and etiquette.
- Writing Effective Subject Lines: Techniques to ensure your emails are opened.
- Emails for Networking: Approaching professionals and mentors in data science/AI.
- Follow-up Emails: Strategies for following up without being intrusive.

2. Report Writing + Presentations

- Structure of a Data Science Report: Elements including abstract, methodology, results, and conclusion.
- Visualizing Data: Incorporating charts, graphs, and other visual tools to enhance comprehension.
- Creating Engaging Presentations: Tips for PowerPoint, storytelling, and engaging your audience.
- Presentation Skills: Delivering your message confidently, handling Q&A sessions.

3. LinkedIn Optimization

- Building a Professional Profile: Key components of a LinkedIn profile for data science/AI professionals.

- Networking Strategies: Connecting with industry professionals and joining relevant groups.
- Content Sharing and Creation: Establishing thought leadership by sharing insights, articles, and engaging with community content.

4. Resume/CV Writing

- Tailoring Your Resume for Data Science/AI: Highlighting relevant skills, projects, and experiences.
- Action Verbs and Quantifiable Achievements: Demonstrating impact in previous roles or projects.
- Design and Layout: Making your resume/CV visually appealing and easy to read.

5. Cover Letter

- Structure of a Cover Letter: Introduction, body, and closing.
- Customizing Your Message: Researching the company and role to personalize content.
- Highlighting Fit and Value: Articulating how your skills and experiences align with the job requirements.

6. Freelancing

- Getting Started with Freelancing: Platforms for data science/AI freelancers, setting up a profile.
- Finding Projects and Clients: Strategies to secure freelance work and build a portfolio.
- Pricing Your Services: Understanding market rates and value-based pricing.
- Client Management: Communicating effectively and managing expectations.

7. Kaggle for Data Science

- Introduction to Kaggle: Overview of the platform, competitions, datasets, and notebooks.
- Participating in Competitions: Tips for success, collaboration, and learning from the community.
- Building a Portfolio: Using Kaggle to showcase your skills and projects to potential employers.

8. GitHub

- Why GitHub for Data Scientists: Importance of version control and code sharing.
- Creating and Managing Repositories: Best practices for organizing and documenting projects.
- Collaborating on Projects: Contributing to open-source projects and collaborating with others.
- GitHub as a Portfolio: Presenting your work and contributions to potential employers.

9. How to Crack Data Analysis Interviews

- Understanding the Interview Process: Types of interviews (technical, behavioral, case studies).
- Preparing for Technical Interviews: Common questions, coding challenges, and statistical questions.
- Behavioral Interview Preparation: Crafting your story, STAR method for responses.
- Mock Interviews: Practicing with peers or mentors to gain confidence.

10. Global Market Understanding

- Data Science/AI Trends: Understanding global trends and emerging technologies.
- Cultural Competence: Working in multicultural teams and serving diverse user bases.
- Regulatory Environment: Overview of data privacy laws and ethical considerations in different regions.

12. Storytelling Using Data

- Principles of Data Storytelling: Crafting narratives that resonate with your audience.
- Visual Narrative Techniques: Using data visualizations effectively in your story.
- Engaging Presentations: Combining data, visuals, and narrative for impactful presentations.

13. Intro to Data Commons

- Understanding Data Commons: Concept, importance, and examples.
- Accessing and Contributing to Data Commons: Guidelines and best practices.
- Leveraging Data Commons: How data scientists can use these resources for research and development.